

BIOS vs UEFI Comparison Table

Aspect	BIOS	UEFI
Definition	Basic Input Output System, an older type of firmware that comes built into the motherboard	Unified Extensible Firmware Interface, a newer firmware that replaced BIOS with more features
Release Era	Introduced in the early 1980s, used for decades	Introduced in the mid 2000s, became common after 2012
Boot Process	Uses Master Boot Record (MBR), reads the first sector of the disk in what is called real mode	Uses GUID Partition Table (GPT), can boot directly in protected or long mode, which is faster
Maximum Disk Support	Supports disks only up to 2 TB	Supports disks larger than 2 TB, technically up to 9.4 zettabytes
Partition Style	MBR only, limited to 4 primary partitions	GPT, supports up to 128 partitions
Interface	Plain text menus, navigated using only the keyboard	Graphical menus, usually supports a mouse, easier to read and use
Security Features	No built in security checks during boot	Has Secure Boot, which blocks unknown or unsigned operating systems from loading
Speed	Slower startup, checks hardware step by step in a fixed order	Faster startup, can check hardware at the same time instead of one by one
Driver Support	Limited, only supports basic hardware drivers during boot	Supports more modern drivers and can run small programs before the OS loads
Network Support	Cannot connect to a network before the OS loads	Some UEFI systems can access network features before the OS loads, useful for remote repair
Compatibility	Works on almost any hardware, even very old machines	Not compatible with very old motherboards, needs newer hardware
Advantages	Simple, reliable, works on legacy systems, easy to understand	Faster boot, supports large drives, has Secure Boot, better interface, more flexible
Disadvantages	Slow boot, 2 TB disk limit, no Secure Boot, harder to use for beginners	More complicated setup, does not work on old hardware, some older operating systems are not supported
Modern Usage	Rarely used now, mostly found in computers built before 2010	Standard on nearly every new computer, required for Windows 11

Why UEFI Has Replaced BIOS

For a long time, **BIOS** was the only firmware that computers used to start up. It did its job, but it was built for a time when computers were much simpler and hard drives were much smaller. As technology moved forward, the limits of BIOS became more obvious. One of the biggest problems was the 2 TB disk size limit. Once storage drives grew past that size, which happens easily today, BIOS simply could not read the rest of the space. This made it a poor fit for modern computers that often come with drives much bigger than that.

Another issue was speed. BIOS checks each piece of hardware one at a time before the operating system can load. This process, called **POST**, takes longer than it needs to. **UEFI** improves on this by allowing hardware checks to happen faster and sometimes at the same time, which cuts down boot time noticeably.

Security is probably the biggest reason companies moved to UEFI. BIOS has no way to check if the software trying to load during startup is safe or not. This makes it easier for certain kinds of malware to hide deep in the system before the operating system even starts. UEFI solves this with a feature called **Secure Boot**. This feature checks the digital signature of the software trying to load, and if it is not from a trusted source, the system will stop and refuse to load it. For businesses and schools that care about protecting their systems, this feature alone makes UEFI a much safer choice.

The way partitions work also changed for the better. BIOS depends on the Master Boot Record, which only allows four primary partitions and cannot handle large drives properly. UEFI uses the **GUID** Partition Table instead, which supports far more partitions and removes the old size limits completely. This gives users and administrators a lot more freedom when setting up storage on a server or a personal computer.

Ease of use matters too. Old BIOS screens are plain, text based, and can only be controlled using a keyboard. This was fine for the time, but modern UEFI screens usually have a graphical layout and can be controlled with a mouse, which feels a lot more familiar to most people today.

Because of all these **improvements**, faster boot times, support for bigger drives, stronger security, and a better interface, most computer manufacturers stopped using **BIOS** and switched fully to **UEFI**. Newer operating systems like Windows 11 even require UEFI to install, which shows how much the industry has moved away from the old system. BIOS still shows up occasionally on very old machines, but for anything built in the last several years, UEFI is now the **standard** almost everyone uses.